Share or Hide? Strategic Disclosure in Congested Low-Altitude Drone Traffic

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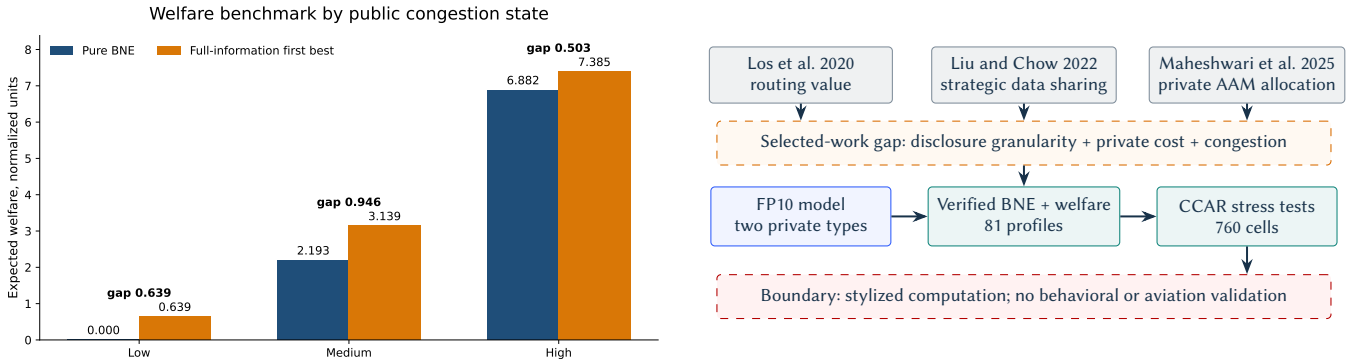


Figure 1: Verified evidence chain. Left: code-recomputed Bayesian-equilibrium (BNE), CCAR, and full-information first-best welfare. Right: a paper-specific readable summary of inherited foundations, FP10's bounded computational contribution, and unresolved validation. Values are normalized stylized quantities, not aviation measurements.

Submission to: COMSCI/ECON 206 — Computational Microeconomics. **Session:** Autumn 2026 Session 1.
Team: FP10. **Instructor:** Professor Luyao Zhang. **Final deadline:** 21 October 2026, 23:00 Asia/Shanghai.

1 Background

Two commercial operators share a congested low-altitude coordination environment. Each knows its private confidentiality-cost type but not the rival's. They choose Minimum, Partial, or Full voluntary commercial disclosure; Minimum still includes the model's assumed safety-information floor. Information sharing can improve routing coordination [2], while competing operators and airspace institutions create strategic and privacy constraints [1, 3].

2 Motivation

The institutional problem is not "maximize disclosure." Operators internalize confidentiality cost but jointly create congestion-dependent coordination benefits. A privately rational equilibrium can therefore miss total-welfare opportunities, while an access incentive can also induce excessive disclosure. The project asks where a candidate rule helps and where it fails, rather than treating one favorable calibration as a theorem.

3 Application Scenario

Operator A and B coordinate commercial flight intent in a capacity-constrained urban setting. Partial may encode a coarse next sector/ETA; Full is richer route/timing detail. Mandatory safety information, emergency missions, enforcement, inaccurate reports, and real traffic control lie outside the action space. Stakeholders include operators, a coordinator, customers, and affected publics. The model is a normalized computational benchmark, not an empirical aviation forecast.

4 Research Question

Shared: How do private confidentiality costs and public congestion shape voluntary disclosure, welfare, and the performance of a congestion-contingent access rule? **Economics** defines incentives and welfare; **computer science** exhaustively enumerates equilibria and falsifying parameter regions; **behavioral science** motivates future tests of comprehension, privacy salience, and fairness—no behavioral population result is claimed.

5 Method

Nature independently draws $c_i \in \{1, 1.6\}$ with probability $1/2$; public congestion is $d \in \{0.3, 0.6, 1\}$. Simultaneous actions are $a_i \in \{M, P, F\}$ with $x = (0, 1, 2)$ and $k = (0, 1, 2.5)$. Utility is

$$u_i = d(7.5)[1 - e^{-0.5(x_i + x_j)}] - c_i k(a_i).$$

A complete pure strategy maps both own types to actions. The solver checks all $9 \times 9 = 81$ strategy pairs for interim best responses; symmetry is not imposed. Welfare is $W = u_A + u_B$. The full-information first best observes realized types and is a reference, not necessarily feasible. CCAR scales only a Minimum discloser's access to enhanced, non-safety coordination benefit by $\alpha_M = 0.7$ at Medium/High congestion. It does not withhold mandatory safety information or establish truthful reporting. The downstream allocation appendix separately compares FCFS with a one-slot second-price auction; it does not turn the disclosure game into an auction.

Artifacts: GitHub repository; Colab 01; Colab 02; Hugging Face Space. Validated source commit: `96a1c3a5276267ff6a039d9a11a3d9c4b2cfce38`.

6 Results

All benchmark states have one pure BNE. Low: (M, M), $W = 0.000$, first best 0.639, gap 0.639. Medium: (P, M) by type, $W = 2.193$, first best 3.139, gap 0.946. High: (P, P), $W = 6.882$, first best 7.385, gap 0.503. Thus the largest benchmark gap is Medium. There, CCAR changes the BNE to (P, P) and underlying welfare to 3.089, leaving gap 0.050. These are code-recomputed normalized expectations. Across 760 deterministic grid cells, diagnostics overlap: 401 no equilibrium-set effect, 334 robust welfare improvements, 18 excessive-disclosure flags, 3 larger-gap cases, 66 multiple-CCAR-BNE cases, and 0 without a pure CCAR BNE. The cells are neither subjects nor probabilities.

7 Intellectual Merits

Los et al. quantify operational sharing value in collaborative vehicle routing [2]; Liu and Chow study efficient/stable transit-operator data sharing [1]; Maheshwari et al. design privacy-preserving AAM coordination [3]. FP10's bounded difference is the exact link among private binary confidentiality types, three disclosure levels, public congestion, complete Bayesian strategies, a realized-type first-best benchmark, and CCAR failure regions. This is a reproducible application/model combination and comparative numerical result—not a general theorem, proof of optimality, or empirical novelty claim. Appendix G expands Figure 1.

8 Practical Impacts

Potential relevance is limited to UN SDG targets 9.1 (sustainable/resilient infrastructure) and 11.2 (safe, accessible, sustainable transport). The demonstrated evidence is only stylized welfare/disclosure computation. Proposed indicators are coordination delay, confidential-data exposure, access disparity, and override/safety incidents relative to an unrestricted-access baseline. Risks include high-cost-operator disadvantage, gaming, and over-disclosure; safeguards are a safety-information floor, auditable eligibility, appeal, privacy review, and suspension on adverse diagnostics. Academia requires reproducibility; industry additionally requires reliability/security; public institutions require legitimacy, equitable access, contestability, and correction. No emissions or real-world safety benefit is claimed.

Future Research Roadmap

Economics: add reporting/participation and implementation cost, with welfare and individual-rationality criteria. Computer science: simulate dynamic heterogeneous multi-agent traffic and privacy-preserving coordination, with scalability and failure-rate criteria. Behavioral science: preregister comprehension/framing choices before any operator inference, subject to ethics review. Han Zhang's suggested independent-coordinator role is a proposed three-party extension, not a completed model. The sequence is current benchmark → incentives → dynamic computation → behavioral/institutional validation → possible field evaluation.

Evolving 2056 Contribution Statement

By 2056, the aspiration is accountable incentive-aware coordination in shared autonomous systems. The current contribution is only the verified stylized comparison above. Official records identify the 2025 prize to Joel Mokyr, Philippe Aghion, and Peter Howitt for explaining innovation-driven economic growth [4], and the 2026 prize to Robert E. Hall and Michael Woodford for microeconomic foundations of macroeconomic stabilisation [5]. These ideas motivate long-horizon mechanism-building grounded in agent behavior; neither prize validates FP10. Progress requires general theory, independent replication, behavioral evidence, and safe field evaluation.

Open Science and SDG Statements

Shortest local reproduction runs the repository's central evidence builder; model data, grids, figures, tests, and manifests are public alongside it. Local notebook execution is recorded; hosted Colab and final deployment parity remain separately labeled. SDG connections are prospective until stakeholder-defined field indicators and a governance process exist.

Author Notes

Full Acknowledgments

Yiqiao Liu thanks Professor Luyao Zhang for the COMSCI/ECON 206 assignment, official templates, and emphasis on evidence-linked interdisciplinary research; Professor Kenneth S. (Ken) Rogerson for his documented role in the course's 7 September 2026 "2056 Nobel and Turing Laureate Program Launch Workshop" context (no direct FP10 paper review is claimed); Feiyu Li (FP3 member 1) for a documented 28 September review of a superseded AI memory-portability version; Han Zhang for the author-supplied suggestion to study an independent third-party oversight extension; and Xinrong Sun for the author-supplied novelty, domain-necessity, congestion-benefit, and CCAR-externality challenges. Feiyu Li's review is retained only as development history, not as validation of the current drone model; exact dates/wording of the latter two records await author confirmation. We acknowledge Tencent's Shanghai office, the Shanghai Science and Technology Museum, and the COMSCI/ECON 206 classroom community as course field-learning/community contexts, without attributing unrecorded FP10 findings to them. Official UN SDG assets are credited to the United Nations. Python, NumPy, Matplotlib, nbclient, Gradio, Chromium/Playwright, the ACM class, and the instructor-supplied templates supported the artifacts. OpenAI Codex assisted the final audit, code verification, drafting, figures, and production; Yiqiao Liu remains responsible for every claim, source, computation, and submission.

Updated Team and Individual Contributions

Team FP10 is a solo project. Yiqiao Liu chose to preserve a three-action Bayesian disclosure game, distinguish mandatory safety information from voluntary commercial detail, interpret full information as an infeasible reference, and test CCAR as a candidate rather than an optimum. Liu authored and verified `src/`, `tests/`, the two notebooks, the Hugging Face/static demo, deterministic result manifests, paper, diagrams, and poster. Verification includes exhaustive 81-profile equilibrium enumeration, welfare arithmetic, 760-cell robustness, belief/tolerance sensitivities, local notebook execution, automated tests, and rendered artifact inspection. The integrated consequence is that computation found failure regions, which narrowed both the mechanism and practical-impact claims and shaped the behavioral/institutional roadmap.

References

[1] Qi Liu and Joseph Y. J. Chow. 2022. Efficient and Stable Data-Sharing in a Public Transit Oligopoly as a Coopetitive Game. *Transportation Research Part B: Methodological* 163 (2022), 64–87. doi:10.1016/j.trb.2022.06.013

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[3] Chinmay Maheshwari, Maria G. Mendoza, Victoria Tuck, Pan-Yang Su, Victor L. Qin, Sanjit A. Seshia, Hamsa Balakrishnan, and Shankar Sastry. 2025. Privacy-Preserving Mechanisms for Coordinating Airspace Usage in Advanced Air Mobility. *ACM Journal on Autonomous Transportation Systems* 2, 4 (2025), 1–34. doi:10.1145/3732290

[4] Nobel Prize Outreach. 2025. Press Release: The Prize in Economic Sciences 2025. NobelPrize.org. https://www.nobelprize.org/prizes/economic-sciences/2025/press-release/ Official prize announcement; accessed 15 October 2026.

[5] Royal Swedish Academy of Sciences. 2026. The Laureates Have Helped Us Smooth Out the Economy's Peaks and Troughs. Prize in Economic Sciences announcement. https://www.kva.se/en/prizes/prize-in-economic-sciences/ Official Academy announcement; accessed 15 October 2026.

[6] William Vickrey. 1961. Counterspeculation, Auctions, and Competitive Sealed Tenders. *The Journal of Finance* 16, 1 (1961), 8–37. doi:10.1111/j.1540-6261.1961.tb02789.x

A Technical Details and Reproducibility

For type-contingent strategies s_i, interim expected utility is

$$EU_i(a \mid t_i, s_j, d) = \sum_{t_j} \Pr(t_j) u_i(a, s_j(t_j); t_i, d).$$

A profile is a pure BNE iff each prescribed action maximizes this expression for both types and both players. The implementation enumerates all 3^2 strategies per player and all 81 pairs, preserving all ties within tolerance 10^{-10}. For each realized type pair the first best enumerates nine action profiles, maximizes $u_A + u_B$, preserves ties, and averages under the prior. The gap is exactly $W_{FB} - W_{BNE}$, evaluated using underlying welfare even when CCAR modifies access utility.

Table 1: Verified normalized primitives and benchmark.

State	BNE (L, H)	W_{BNE}	W_{FB}	Gap
Low ($d = .3$)	(M, M)	0.000	0.639	0.639
Medium ($d = .6$)	(P, M)	2.193	3.139	0.946
High ($d = 1$)	(P, P)	6.882	7.385	0.503

CCAR input is congestion, own disclosure action, and α_M. Its output is an enhanced-information access factor: 1 at Low; α_M for Minimum at Medium/High; 1 for Partial/Full. It assumes observable/verifiable action, contains no report-quality model, has no administration/enforcement cost, and does not establish voluntary participation or incentive compatibility. At Medium and $\alpha_M = .7$, BNE is (P, P) and underlying welfare is 3.089. The robustness grid is $2 \times 20 \times 19 = 760$ deterministic cells over Medium/High, $c_H = 1.1, \ldots, 3.0$, and $\alpha_M = .1, \ldots, 1.0$. Categories overlap.

Reproduction: create the specified Python environment, run `python scripts/build_final_evidence.py`, then the four test commands in the README. No random seed enters the equilibrium results. The allocation simulation uses seed 206. Environment captured during evidence generation: Python 3.13.7, NumPy 2.3.5, Matplotlib 3.10.8. Known limits: two symmetric operators; independent binary types; three discrete actions; exogenous congestion; pure strategies; uncalibrated normalized primitives; no report accuracy, implementation cost, mixed equilibrium, dynamics, or empirical sample. Validated source commit: `96a1c3a5276267ff6a039d9a11a3d9c4b2cfce38`.

A.1 AI Use Disclosure

OpenAI Codex (GPT-5-family coding agent; exact backend build not exposed) was used on 15 October 2026 for repository/requirements audit, computation and test verification, source checking, academic editing, diagrams, paper/poster production, and release validation. The material prompt is the author's final-project master instruction retained in the Codex

task record. Accepted work includes the central evidence builder, bounded literature comparison, failure-case reporting, release documents, and layout drafts. Rejected/qualified suggestions include absolute novelty, universal welfare improvement, empirical aviation or behavior claims, fabricated human activity, and successful deployment without evidence. Automated checks are logged in the release documentation. Final upload, historical-credit, and scientific acceptance remain human responsibilities; Yiqiao Liu bears responsibility.

B Cumulative Intellectual Development

Verified continuity begins with the tracked PS2 drone-disclosure repository at commit `6baa4107`. The final revision preserves its player/type/action space, welfare benchmark, CCAR, allocation exercise, notebooks, and decision lab. The 15 October audit added a central evidence source, belief/tolerance sensitivities, explicit category overlap, three closest-paper verification, and diagrams. A recovered 28 September review by Feiyu Li concerned a superseded AI memory-portability version; its multiplicity, behavioral-evidence, and reproducibility lessons are retained without reclassifying it as current-model review. Han Zhang's supplied three-party suggestion became a roadmap test rather than a claimed feature. Xinrong Sun's supplied objections caused four consequential revisions: narrow novelty; label drone claims as application interpretation; expose congestion benefit as an assumption; and foreground CCAR no-effect/adverse regions. Week 1, Week 2, PS1, October 12, October 14, Ed, email, and handwritten records are not present in the repository; their detailed historical content remains pending rather than inferred.

C Open Source and Three Lens Inspiration

Game theory supplies Bayesian strategies and best-response conditions; social choice distinguishes individual utility from aggregate welfare and exposes omitted distributional criteria; mechanism design changes enhanced-access payoffs and tests implementation trade-offs. Python/NumPy implement exhaustive deterministic enumeration; Matplotlib produces reproducible figures; nbclient supports clean local notebook execution; Gradio/static HTML provide an illustrative decision interface. Official UN icons remain UN property and are used for academic communication with source attribution. The ACM/instructor LaTeX and A0 templates define delivery structure. The repository's own license governs project reuse; third-party components retain their licenses.

D Feedback Response and Revision Record

Table 2: Consequential feedback responses.

Source	Challenge	Decision and revision	Verification / open issue
Han Zhang	Three-party oversight	INVESTIGATE: roadmap adds independent coordinator	Current solver remains two-player; original record needed
Xinrong Sun	Los et al. novelty	REVISE: exact three-paper comparison; bounded contribution	DOI/publisher metadata checked; intended citation needs confirmation
Xinrong Sun	Drone necessity / congestion assumption	REVISE: abstraction boundary and sensitivity	Code/model and grid inspected; no calibration
Xinrong Sun	CCAR externalities	REVISE: report no-effect, excessive, larger-gap, multiplicity	760 cells recomputed
Feiyu Li (28 Sept.)	Superseded-topic multiplicity, observed choice, reproducibility	RETAIN LESSONS: multiplicity and behavioral boundaries	Not direct review of current drone model
Official assignment	Final artifacts and structure	REVISE: official paper/poster template and release checks	Build record in Appendix G

The repository's final-documentation folder contains the complete machine-readable feedback register. Direct FP10 feedback from Ken Rogerson is not claimed.

E Structured Abstract and Cumulative Proposal Map

Background and motivation. Two commercial drone operators choose how much non-safety flight information to disclose while facing private confidentiality costs and public congestion; individually rational choices may miss joint coordination value. **Application and question.** In a stylized urban low-altitude setting, when do equilibrium disclosure and welfare depart from a realized-type first best, and when does a congestion-contingent access rule improve or worsen that gap? **Method.** The project exhaustively enumerates all 81 pairs of complete pure strategies, computes Bayesian equilibria and first-best welfare, and stress-tests CCAR across 760 deterministic parameter cells. **Results.** Baseline welfare gaps are 0.639, 0.946, and 0.503 at Low, Medium, and High congestion; the worked Medium CCAR case reduces the gap to 0.050, while the grid also finds no-effect, excessive-disclosure, larger-gap, and multiple-equilibrium regions. **Intellectual merits.** The contribution is a reproducible, bounded combination of disclosure granularity, private cost, public congestion, welfare benchmarking, and explicit failure mapping—not a general theorem. **Practical impacts.** Possible SDG 9.1/11.2 relevance remains prospective and requires safety, privacy, distributional, behavioral, and field validation.

Table 3: Eight required parts and consequential final revision.

Part	Current statement and revision evidence
Background	Competing operators choose non-safety disclosure under private confidentiality cost and public congestion; three closest papers verified.
Motivation	Private incentives can miss aggregate welfare, but more disclosure is not always better.
Application Scenario	Stylized low-altitude coordination; mandatory safety, emergencies, and empirical operations excluded.
Research Question	Conditions for equilibrium disclosure, welfare gaps, and CCAR improvement/failure.
Method	Complete Bayesian strategies, exhaustive pure-BNE enumeration, first best, deterministic robustness.
Results	Benchmark reproduced; Medium gap 0.946 $\rightarrow$ 0.050 under the example; adverse/no-effect regions reported.
Intellectual Merits	Bounded new model/application combination and comparative computation, not a theorem.
Practical Impacts	Prospective SDG 9.1/11.2 connection with indicators, risks, safeguards, and evidence labels.

F Retained PS2 Auction Artifact and Symposium Record

The upstream disclosure game is not an auction. The downstream exercise allocates one residual commercial priority slot; emergency/public-safety missions are excluded from payment-based access. Participants have private scalar values and submit sealed bids. FCFS allocates to the first eligible request, charges zero, and stops. The alternative single-unit second-price auction allocates to the highest bid, charges the second-highest bid, breaks ties by operator ID, and stops. Under the independent-private-values benchmark, truthful bidding is weakly dominant [6]. For values $(8, 5, 3, 2)$ and arrival B,C,A,D, FCFS chooses B (efficiency .625); second price chooses A, payment 5, utility 3 (efficiency 1). In 10,000 seed-206 draws, mean FCFS efficiency is .622308, second-price efficiency 1, and auction revenue 5.99207. This illustrates scarcity allocation only; it does not establish aviation legitimacy, distributional fairness, willingness/ability-to-pay equivalence, or CCAR performance. A genuine 28 September review by Feiyu Li was recovered, but it concerns the superseded AI memory-portability version. A separate symposium record for the current drone topic was not recovered and awaits the author.

G Final Week Evidence and Final Release Audit

Evidence status. Feiyu Li's dated 28 September superseded-topic review is recovered and bounded above. October 12 Ed post, duet thread, Category I/II/III posts, and October 14 peer replies, roundtable, Ken Rogerson exchange, and handwritten exit records were not recovered. They remain structured placeholders in `docs/final/final_week_evidence.md`. Known Han Zhang/Xinrong Sun summaries are used only as author-supplied summaries. No fictional transcript, date, or participant is supplied.

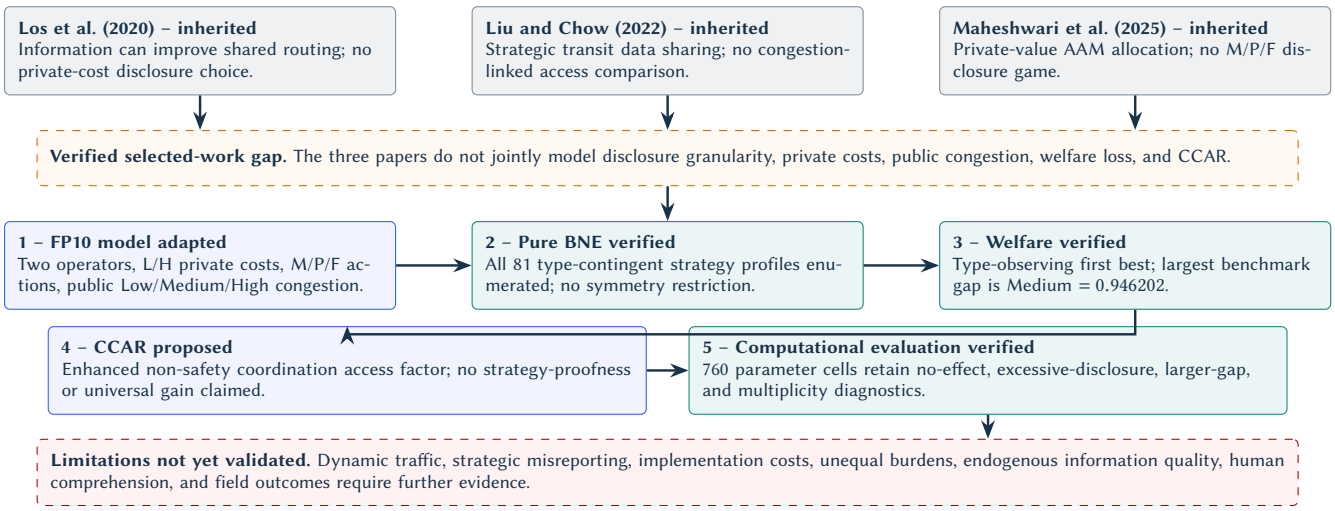


Figure 2: Expanded Intellectual Merits chain. Gray-blue marks inherited literature; blue and green distinguish the adapted model, proposed CCAR, and verified computation; dashed orange/red boxes mark the bounded gap and unresolved limitations.

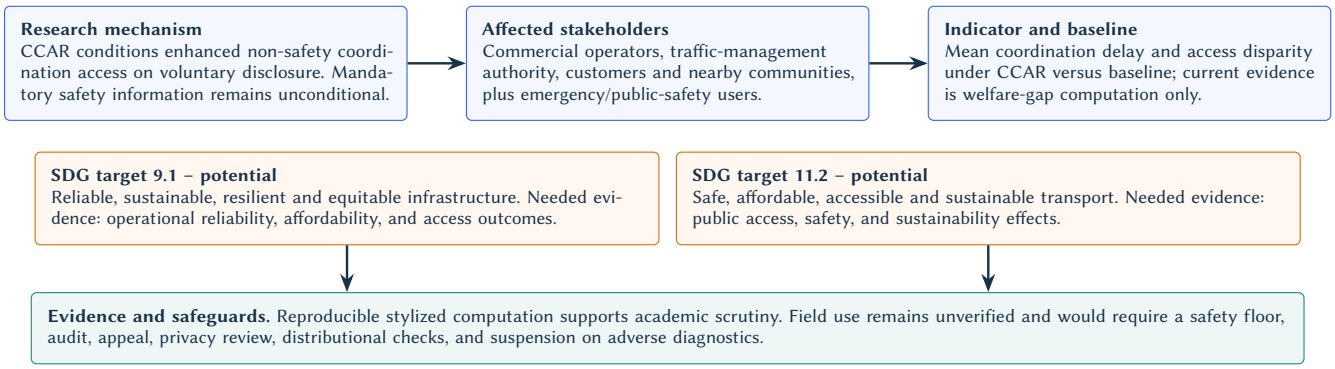


Figure 3: Prospective impact map. SDG targets are interpretive links; only stylized computational evidence is demonstrated.

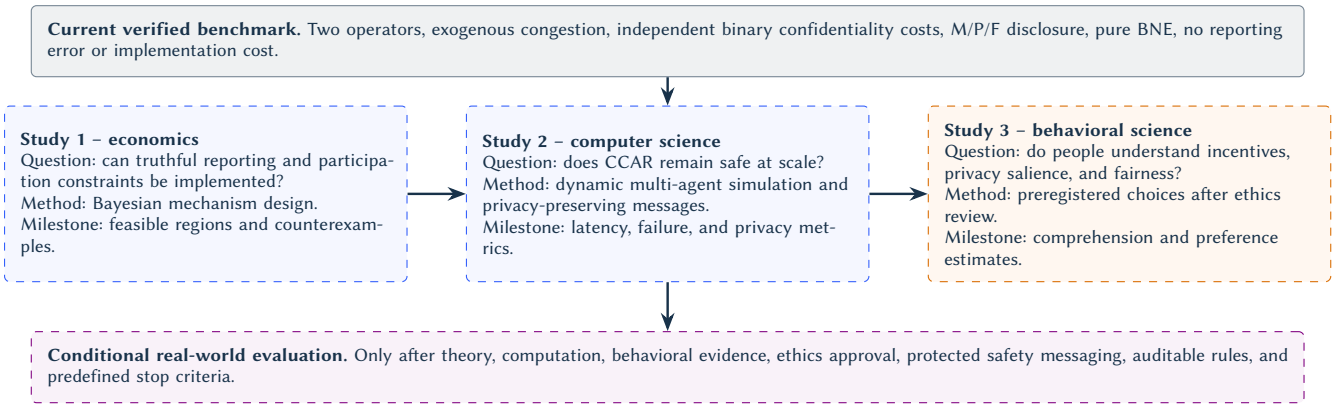


Figure 4: Future roadmap linking current limitations to economics, computer-science, behavioral, and institutional tests.

Nobel comparison. Official records identify the 2025 prize to Joel Mokyr, Philippe Aghion, and Peter Howitt for innovation-driven growth [4], and the 2026 prize jointly to Robert E. Hall and Michael Woodford for microeconomic foundations of macroeconomic stabilisation [5]. The comparison motivates linking agent-level incentives to long-run institutional outcomes; it does not validate FP10 or imply prize merit.

Release audit. Full checks and hashes are in the release documentation. Validated source commit: `96a1c3a5276267ff6a039d9a11a3d9c4b2cfce38`. Yiqiao Liu must personally confirm the regenerated files and complete the Canvas upload.